

Moneta Reporting Engine - Detailed Technical Design

1. Architecture Principles

- Clean Architecture dependency rule
- Core independent from frameworks
- Ports & Adapters model
- CQRS Light Implementation

2. Core Layer Design

- ReportOrchestrator
- ReportProvider interface
- RepositoryPort & AuditPort
- Domain Models and DTO mapping

3. Pluggable Reports Layer

- Each UDM packaged as independent module/JAR
- Implements ReportProvider
- Auto-discovered via Spring

4. Infrastructure Layer

- JPA/JDBC adapters
- Multi-datasource support
- Audit adapter (Splunk/DB)
- Security integration

5. Data Flow Execution

- API receives category
- Factory selects provider
- Provider builds query
- Repository adapter executes query
- Formatter builds paginated response
- Async audit event published

6. Performance Engineering

- Streaming result sets
- Index optimization
- Connection pooling (HikariCP)
- Response compression
- Caching layer

7. Scalability Model

- Horizontal scaling
- Stateless API
- Externalized cache
- Async heavy job execution

8. Observability

- Micrometer metrics
- Distributed tracing
- Structured logging
- Health checks

9. Security Model

- Token-based authentication
- Role-based access
- Row-level security (optional)

10. Deployment Architecture

- Docker container
- Private Cloud
- CI/CD integration
- Environment separation (DEV/SIT/UAT/PROD)